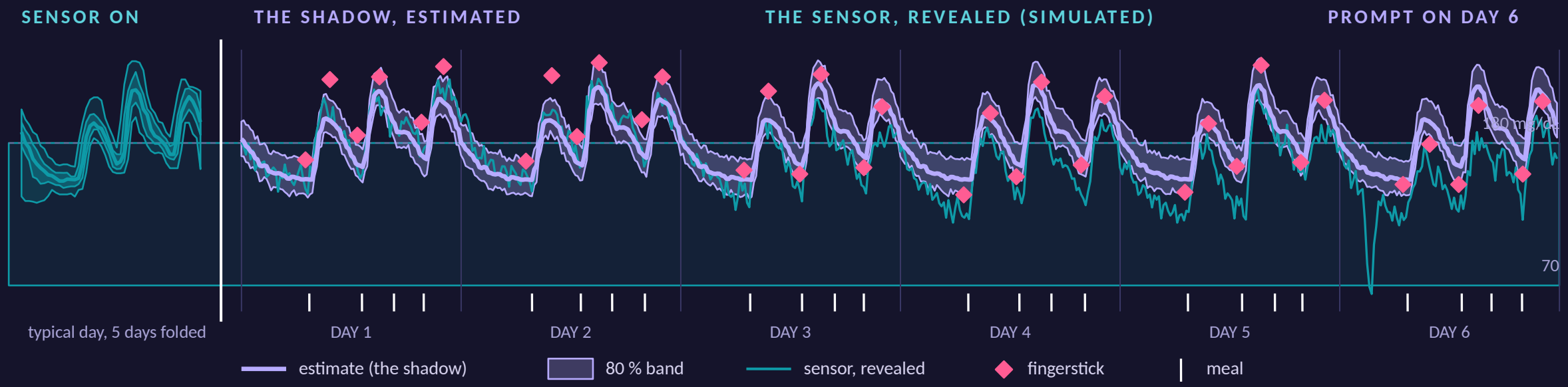
Chhaya: the diabetes digital twin that keeps going when the sensor is off

It fuses a patient's health record with a wearable glucose sensor, draws an honest band around every estimate, says when its report may be out of date, and proves every claim on patients it never saw.



Mrs. R. is a synthetic demo patient and the sensor shown is simulated. Research prototype, not a medical device.

One sensor wear, then months of decisions on an ageing report

101 million

people in India live with diabetes

ICMR-INDIAB national study, 2023

₹4,400–5,100

for one 14-day glucose-sensor wear

Indian retail listings, October 2026

26 sensors

a year to wear one continuously: over ₹1.1 lakh

26 times the lowest listed price

THE SCENARIO

14 days of sensor data

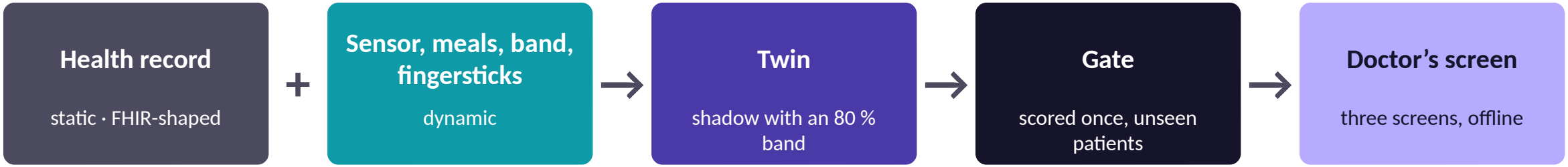
Months of treatment decisions made from that one report

?

Nobody tells the doctor how old that report has become.

Sources: ICMR-INDIAB national cross-sectional study, The Lancet Diabetes & Endocrinology, 2023 (101 million, 11.4 %). Retail listings checked on 11 October 2026 (Truemeds; Amazon India price history); prices vary by model and seller.

Two streams fused into one personal twin, checked against the brief



1	**Static stream**	A FHIR-shaped health record: diagnosis, duration, HbA1c, fasting glucose, BMI, medications.
2	**Dynamic stream**	CGM sensor wear, meal log, wrist-band activity and fingersticks.
3	**Fusion**	The record sets the prior; the sensor wear is the evidence (MAP fit, Laplace ensemble).
4	**Adverse events predicted**	Post-meal excursion (AUROC 0.82 with the sensor, 0.71 without) and a stale-report prompt (AUROC 0.82).
5	**Doctor dashboard**	One panel per stream plus the shadow; runs offline; live demo online.

A physiology-based twin, fitted to one wear and always drawn with its uncertainty

1 Record
Fasting glucose sets the prior

→

2 Wear
MAP fit from six starts

→

3 Ensemble
Laplace posterior, 200 members

→

4 Simulate
E-DES, RK4, JAX float64

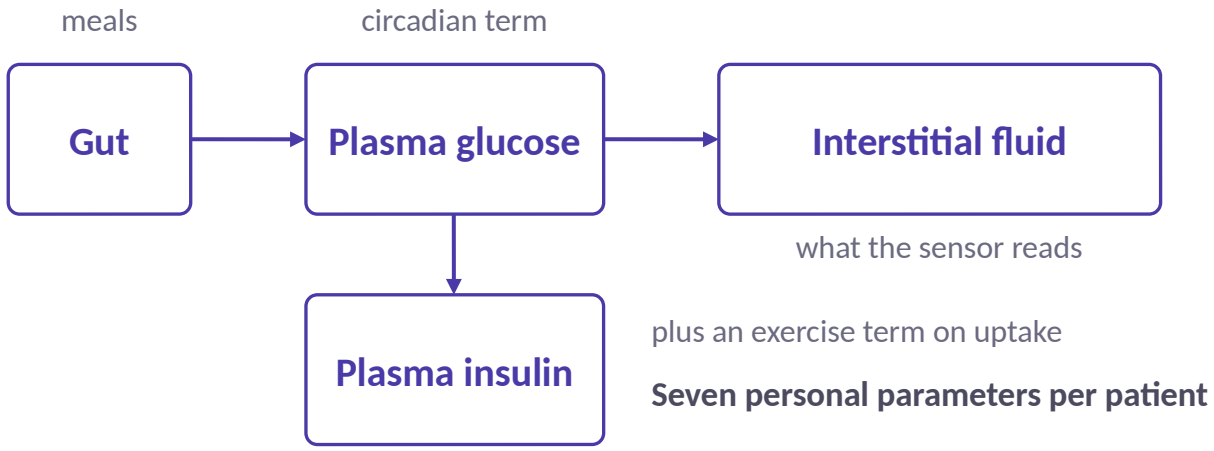
→

5 Blend and nudge
Half own day; fingerstick fade 120 min

→

6 Output
Shadow, 80 % band, CUSUM prompt

THE PHYSIOLOGY: E-DES, MAAS ET AL. 2015



METHOD FACTS

- **No neural network, no pretrained model, no large language model**
- Seven personal parameters fitted per patient from one wear
- Fingerstick filter fades over 120 minutes; CUSUM raises the drift prompt
- Python 3.12 · JAX · SciPy · scikit-learn · FHIR-shaped record

The doctor sees one panel per stream, plus the shadow



From the health record

The record sets the twin's prior. With sensor data, the sensor dominates.

Meal excursion history

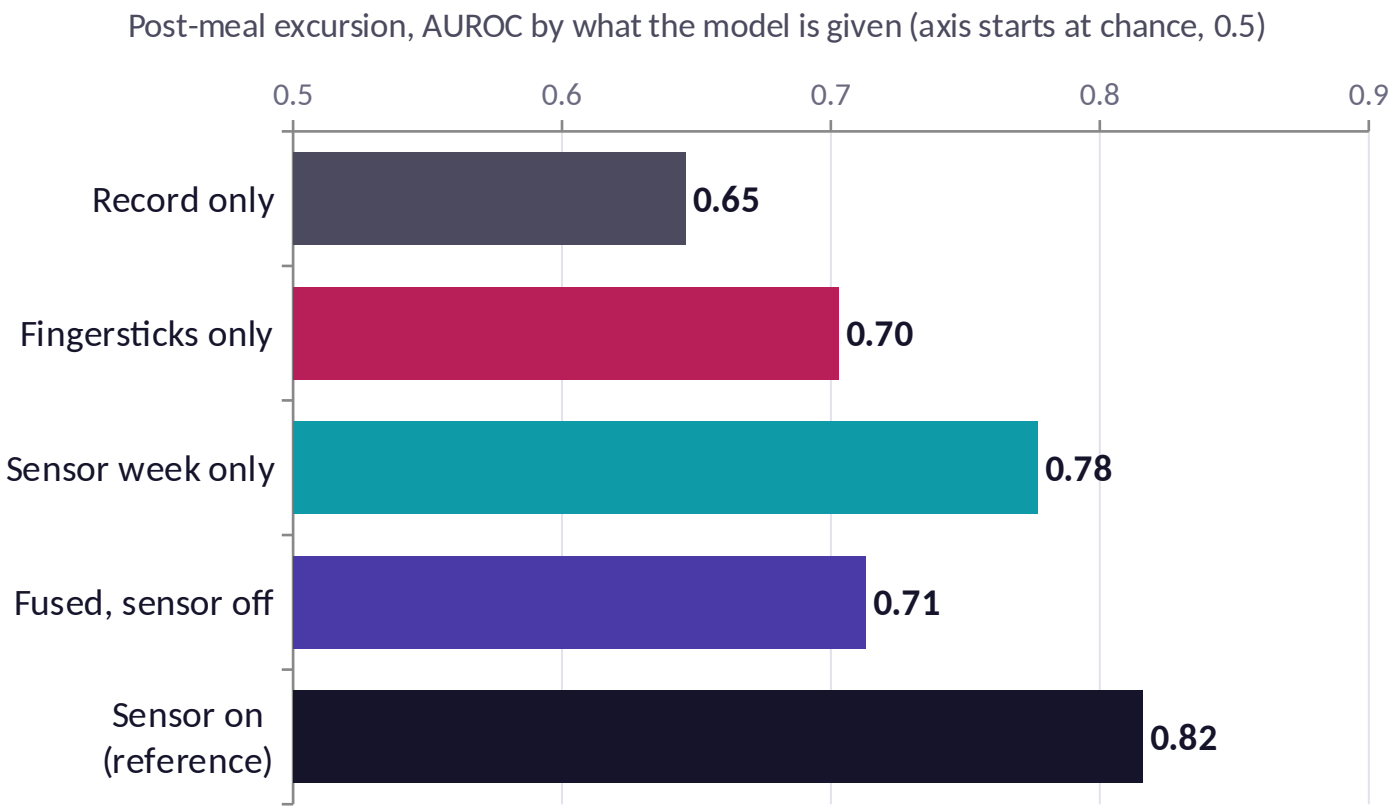
How often each meal type was followed by a reading above 180 mg/dL within two hours during the wear. A description, not a forecast.

Reveal

Hide the sensor, estimate, then slide the real trace over the shadow and see where it was right and where it missed.

Try it: chhaya-digital-twin.vercel.app

With the sensor on, the spike is predictable; with it off, more streams do not help



47 held-out ShanghaiT2DM patients, 1,205 meals, 39 % followed by an excursion. Event: two consecutive readings above 180 mg/dL within 2 hours of a meal start. Exploratory fusion benchmark: 1,436 meals, 49 development patients, patient-grouped cross-validation.

0.82 · 0.76

AUROC · AUPRC with a live sensor

0.71 · 0.60

AUROC · AUPRC, fused, sensor off

8.9

false alerts per patient-week, so no alert is shown

Four other fusion strategies (equal-weight, stacked, recalibrated, stronger regularisation) were tried on development patients: none beat a single stream.

Every claim crossed a gate: bars written first, scored once on unseen patients



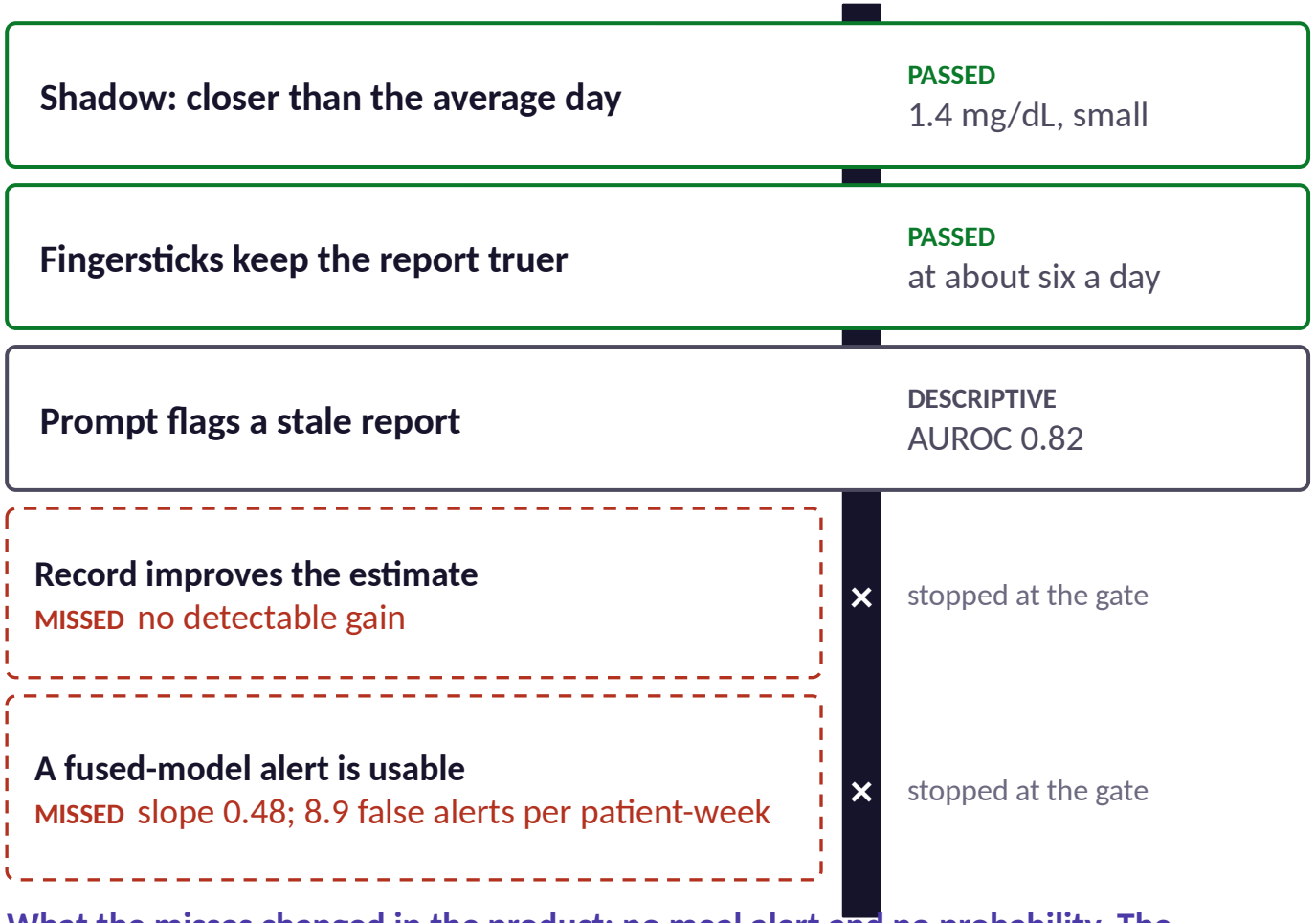
5 passed · 4 missed · all nine published

Half of every cohort tunes the model; the other half is scored once.

The confirmatory command needs --confirm and refuses a second run.

95 % intervals come from a bootstrap over patients (2,000 resamples).

Every claim on the Evidence screen is tested word for word against a decision record.



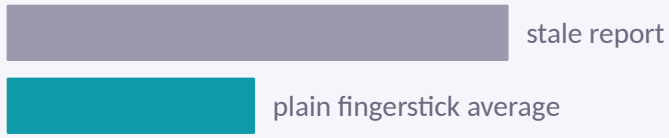
What the misses changed in the product: no meal alert and no probability. The dashboard shows the patient's own excursion history instead.

Three results a clinician can use, each with its conditions

A FINGERSTICK-ANCHORED REPORT

13.0 → 6.4

mg/dL median error



Error of the next days' mean sensor glucose, in hindsight.

29 held-out patients, supervised care, about six fingersticks a day. At one a day the gain was 0.3 mg/dL.

WHICH LOWS NOT TO TRUST

8 of 64

sensor lows confirmed by a fingerstick



None of the 9 night-time lows was confirmed. Unconfirmed is not the same as false.

Readings below 70 mg/dL with a fingerstick within 10 minutes. Descriptive, no bar.

A PROMPT, NOT A FINDING

0.82

AUROC for a drifted sensor report (0.63 to 0.98)



7 of 11 drifted recordings raised it



2 of 21 stable recordings raised it

A running sum of fingerstick surprises. A plain comparison does as well (0.82).

32 held-out recordings, supervised care.

Honest limits, and what Indian validation would take

LIMITS

No Indian patient data. The evidence is a supervised-care cohort (ShanghaiT2DM) and a free-living cohort of mixed glycaemic status (CGMacros).

Small effects. The shadow is 1.4 mg/dL closer than the patient's own average day; the physiology alone beats it in only 60 % of held-out patients.

Conditional gains. The fingerstick result holds at about six a day in supervised care.

Research prototype. No dose advice, no low-glucose estimates, no alarms.

ROADMAP

1

Now

Open-data prototype: nine pre-registered bars, five passed, every number reproducible.

2

Next

An Indian clinic cohort, outpatient fingerstick frequencies of one or two a day, bars written before the run.

3

Then

A prospective check that the stale-report prompt changes a clinical decision.

Every number is a command

Chhaya means shadow. When the sensor goes, the twin stays, and it says when to stop trusting it.

LIVE DEMO

chhaya-digital-twin.vercel.app

CODE

github.com/AdityaChandel11/DigitalTwinHack

REPRODUCE

```
uv sync
uv run python -m chhaya.dashboard
uv run pytest -q
```

400+ tests · 15 decision records · pre-registered bars · code MIT, data CC BY 4.0 and CC BY-NC-SA 4.0

Scan to open the demo

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